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Explainable Artificial Intelligence

Comprehensive Study Notes

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Unit 1

Foundations and Taxonomy of XAI

Required Contact Hours: **9**

• What is Explainability — drivers (regulation, safety, science)

• Definitions — interpretability, transparency, simulatability

• Doshi-Velez/Kim evaluation framework

• Intrinsic models — Linear, GAM, EBM, decision trees

• Taxonomy — local/global, intrinsic/post-hoc, model-agnostic/specific

The chapters that follow expand each topic above with theory, worked examples, reading lists, hands-on lab guidance, and end-of-chapter self-assessment questions.

XAI Module 1 — Foundations: What Is Explainability?

> ***Goal***: *Get the vocabulary right. Most XAI confusion comes from conflating interpretability, explainability, transparency, justification, and fairness. They are not synonyms.*

1.1 Why XAI exists as a field

Three forcing functions:

1. **Regulation** — GDPR Article 22 (right to explanation), EU AI Act high-risk categories, US Algorithmic Accountability Act drafts, India's DPDP Act.
2. **Safety** — deployed ML systems fail in non-obvious ways (medical misdiagnosis, biased lending, hallucination in LLMs).
3. **Science** — understanding why a model works is itself a research goal (Distill.pub, mechanistic interpretability).

DARPA's XAI Program (2017–2021) crystallized the field. Two major surveys: Adadi & Berrada (2018), Guidotti et al. (2018).

1.2 Definitions you must be able to recite

Term	Definition (working)
Interpretability	The degree to which a human can *understand the cause* of a decision. (Miller 2017)
Explainability	The degree to which a system can *produce explanations* — often via post-hoc methods.
Transparency	The model's mechanism is itself inspectable (linear models, small decision trees).
Simulatability	A human can mentally simulate the model's full computation.
Decomposability	Each part of the model has an intuitive meaning.
Algorithmic transparency	Understanding of how the learning algorithm produces a model.
Justification	Reasons that *defend* a decision, not necessarily its true cause.
Causality	The actual mechanism producing the output. (Pearl)

Critical: an "explanation" is not the same as the *cause*. LIME tells you what would have changed the output locally; it doesn't tell you the model's internal mechanism.

1.3 Lipton's framework (Lipton, 2016, "The Mythos of Model Interpretability")

A foundational paper. Lipton split "interpretability" into:

Properties of models (transparency)

- **Simulatability** — a person can mentally run the model.

- **Decomposability** — each parameter / node has a meaning.
- **Algorithmic transparency** — convergence is understood.

Properties of explanations (post-hoc)

- Text explanations
- Visualizations
- Local explanations
- Explanation by example

This split is the canonical taxonomy. Use it.

1.4 Doshi-Velez & Kim (2017) — evaluation framework

Level	What's evaluated	Cost
Application-grounded	Real users on real task	High
Human-grounded	Real users on simpler proxy task	Medium
Functionally-grounded	Formal definition of explanation quality	Low

Most papers operate at the *functionally-grounded* level (e.g., faithfulness scores). Papers that do application-grounded evaluation are scarce and disproportionately influential.

1.5 Who needs an explanation, and why?

Stakeholder	Need
End user	Understand a single decision affecting them
Domain expert (doctor, judge)	Validate model reasoning against domain knowledge
Developer / data scientist	Debug model, audit failures
Regulator / auditor	Establish accountability, check fairness
Researcher	Understand emergent behavior, science

Different stakeholders need *different* explanations. A SHAP plot is useless to a patient. A saliency map is useless to a regulator.

This is the **alignment-to-purpose** problem.

1.6 The accuracy–interpretability tradeoff (myth?)

The folk claim: more complex models → better accuracy but worse interpretability.

Cynthia Rudin's counter (2019, *Nature MI*): *"Stop explaining black-box ML models for high-stakes decisions and* *use interpretable models instead."* On many tabular problems, well-tuned interpretable models (sparse logistic regression, EBMs, falling-rule lists) match black-box accuracy.

The tradeoff is empirically narrower than usually assumed, but **does exist for high-dimensional perceptual tasks** (vision, speech, language).

1.7 Interpretable model families (intrinsic)

Model	Why interpretable
Linear / logistic regression	Coefficients are weights
Generalized Additive Models (GAMs)	Each feature → its own learned curve
Explainable Boosting Machines (EBMs)	GAM + pairwise interactions, often as accurate as XGBoost
Decision trees (small)	Path through tree = rule
Decision rules / RIPPER / SLIM	If-then rules
Falling Rule Lists	Ordered if-then with monotone risk
Risk scores (e.g., 2HELPS2B for seizures)	Integer-coefficient scoring
Generalized linear rule models	Convex combinations of rules
Prototype networks	Predictions justified by similar training examples

A serious XAI engineer **tries an interpretable model first**, then escalates only if needed.

1.8 Post-hoc methods (preview)

For complex models where intrinsic interpretability is unavailable, post-hoc methods explain *after the fact*.

Method	Output
Feature attribution (LIME, SHAP, IG, gradient×input)	Importance score per feature
Saliency maps	Pixel importance per image
Concept-based (TCAV)	Importance of human concept
Counterfactuals	"What would change the prediction?"
Example-based	Nearest training examples
Surrogate models	A simpler model fit to the black box
Mechanistic	Direct circuit analysis (mech interp)

Detailed coverage in Module 3 onwards.

1.9 The "right to explanation"

GDPR's Article 22 grants the right to "meaningful information about the logic involved" in automated decisions. Legal scholars debate whether this is a *right to explanation* or merely a *right to be informed*.

EU AI Act (2024) adds explicit transparency obligations for high-risk AI systems. India's DPDP Act, Canada's AIDA, US state laws all moving similar directions.

This pushes XAI from research curiosity to **compliance requirement**.

1.10 Common pitfalls (memorize)

1. **Confusing correlation explanations with causal ones**. SHAP tells you what the model uses, not what's true in the world.
2. **Trusting explanations of an unfit model**. If accuracy is poor, no explanation is meaningful.
3. **Aggregating local explanations into "global" claims** — disagreement is common (Krishna et al. 2022, "The Disagreement Problem").
4. **Using explanations as fairness audits**. Feature importance ≠ fairness; you need fairness metrics directly.
5. **Optimizing explanation methods for human-likeable outputs** rather than faithfulness.

1.11 Reading list

- Lipton — *The Mythos of Model Interpretability*, ICML WHI 2016.
- Doshi-Velez & Kim — *Towards a Rigorous Science of Interpretable ML*, 2017.
- Miller — *Explanation in Artificial Intelligence: Insights from the Social Sciences*, AI Journal 2019.
- Rudin — *Stop Explaining Black Box ML Models*, Nature MI 2019.
- Adadi & Berrada — *Peeking Inside the Black-Box*, IEEE Access 2018.
- Guidotti et al. — *A Survey of Methods for Explaining Black Box Models*, CSUR 2018.
- Molnar — *Interpretable Machine Learning* (free online book), Ch 1–3.

1.12 Lab (1 hr)

1. Fit a logistic regression and an XGBoost on the same tabular dataset (UCI Adult is fine).
2. Tabulate accuracy + a 3-sentence "explanation" you would give to (a) a regulator, (b) an end user, (c) a developer for each model.
3. Note which audience you struggled to serve.

1.13 Quiz

1. Distinguish *interpretability* and *explainability* in your own words.
2. State Rudin's argument against post-hoc explanations for high-stakes decisions.
3. List the three Doshi-Velez/Kim evaluation levels in order of rigor.
4. Why is "right to explanation" legally contested?
5. Give one example each of: an intrinsically interpretable model, a post-hoc local method, a post-hoc global method.

XAI Module 2 — Taxonomy of XAI Methods

> ***Goal***: *Build a mental map that lets you place any new XAI paper within 30 seconds.*

2.1 The four-axis taxonomy

Every XAI method can be classified along four axes:

```
                 Scope                  Stage                 Model dependency            Output for
                   │                      │                          │                          │
     ┌─────────────┴─────────────┐   ┌────┴─────────┐   ┌───────────┴────────┐   ┌─────────────┴─────
     │                           │   │              │   │                    │   │
  Local                    Global  Intrinsic    Post-hoc  Model-specific   Model-agnostic      Visu
                                                                                                Exam
```

A typical paper introduces a method that is e.g. **local + post-hoc + model-agnostic + visual** (LIME for images).

2.2 Scope: Local vs Global

Local explanations

Why did the model predict ŷ for *this* input `x`?

- **LIME** — locally fit a linear model around `x`.
- **SHAP (local)** — Shapley values for the single prediction.
- **Integrated Gradients** — attribution along path from baseline to `x`.
- **Counterfactuals** — minimum change to `x` that flips the prediction.

Global explanations

How does the model behave *overall*?

- **PDP / ICE / ALE** — marginal effect of a feature.
- **Feature importance** (permutation, gain-based).
- **Surrogate models** — fit a decision tree to the black box's predictions.
- **Concept Activation Vectors (TCAV)** — sensitivity to human-defined concepts.
- **Mechanistic interpretability** — circuit-level reverse-engineering.

Cohort explanations

Subsets between local and global — e.g., "Why does the model behave differently for women than men?" Tools: SliceFinder, Aequitas, FairLearn.

2.3 Stage: Intrinsic vs Post-hoc

Type	Description	Example
Intrinsic	Model is interpretable by construction	Linear, GAM, EBM, small tree, prototype net
Post-hoc	Explain a trained black-box model	LIME, SHAP, Grad-CAM

Hybrid: Self-explaining neural networks (Alvarez-Melis & Jaakkola, 2018), ProtoPNet, Concept Bottleneck Models — *trained* to produce explanations, sit between the two.

2.4 Model dependency

Type	Works on
Model-agnostic	Any model (only needs predict()) — LIME, SHAP, PDP, permutation importance
Model-specific	Exploits internals — Grad-CAM (CNNs), Attention Rollout (Transformers), TreeSHAP (trees)

Model-specific methods are usually **faster and more faithful** to the model's actual computation; model-agnostic ones are **more portable**.

2.5 Output form

Form	When useful
Feature attribution (numeric per feature)	Tabular, NLP, CV
Saliency map (per pixel)	Vision
Attention heatmap	Transformer-based models
Concept score	High-level semantic reasoning
Counterfactual example	"What would change the prediction?"
Prototype / nearest example	Case-based reasoning
Natural-language rationale	LLMs, multimodal
Decision rule / tree	High-trust auditing

2.6 The disagreement problem (Krishna et al., 2022)

When you run multiple explanation methods on the same model and prediction, they often disagree. Practical consequence: an analyst can pick the explanation that supports their narrative.

Implication: never rely on a single XAI method. Triangulate.

2.7 Evaluating an XAI method — three lenses

1. **Faithfulness** — does the explanation reflect what the model actually does?
    - Perturbation-based tests (delete top-k features → measure accuracy drop)
    - Sanity checks (Adebayo et al. 2018 — saliency maps that survive random-weight randomization are not faithful)
2. **Plausibility / human alignment** — does it make sense to humans?
    - User studies
    - Agreement with human-annotated rationales (ERASER benchmark)
3. **Robustness** — does the explanation change under small input perturbations?
    - Adversarial XAI papers (Slack et al. 2020, "Fooling LIME and SHAP")

Detailed in Module 10.

2.8 The XAI method tree (cheat sheet)

```
INTRINSIC
├─ Linear / Logistic / Ridge / Lasso
├─ GAM / EBM
├─ Decision trees (small)
├─ Decision rules (RIPPER, SLIM)
├─ Risk scores (integer-coef)
├─ Prototype networks (ProtoPNet, this-looks-like-that)
└─ Concept Bottleneck Models

POST-HOC
├─ Local
│  ├─ LIME
│  ├─ SHAP (KernelSHAP, TreeSHAP, DeepSHAP)
│  ├─ Anchors
│  ├─ Counterfactuals (Wachter, DiCE, FACE, MACE, GeCo)
│  ├─ Gradient methods
│  │  ├─ Vanilla saliency
│  │  ├─ Integrated Gradients
│  │  ├─ SmoothGrad
│  │  ├─ Guided Backprop
│  │  └─ Grad-CAM / Grad-CAM++
│  └─ Influence functions
└─ Global
   ├─ PDP / ICE / ALE
   ├─ Permutation importance
   ├─ Surrogate models (global tree, distillation)
   ├─ Concept-based (TCAV, ACE, CRAFT)
   ├─ Mechanistic interpretability (probing, circuits)
   └─ Functional ANOVA decomposition

EVALUATION
├─ Faithfulness (deletion / insertion, sufficiency, comprehensiveness)
├─ Sanity checks (model/data randomization)
├─ Stability / robustness
├─ Plausibility (human studies, ERASER)
└─ Run-time / computational cost
```

Print this. Stick it on your wall.

2.9 Reading list

- Adadi & Berrada — *Peeking Inside the Black-Box*, IEEE Access 2018.
- Guidotti et al. — *A Survey of Methods for Explaining Black Box Models*, CSUR 2018.
- Arrieta et al. — *Explainable AI (XAI): Concepts, Taxonomies, Opportunities and Challenges*, Information Fusion 2020.
- Krishna et al. — *The Disagreement Problem in Explainable ML*, 2022.
- Adebayo et al. — *Sanity Checks for Saliency Maps*, NeurIPS 2018.

2.10 Lab (1 hr)

Take last lab's XGBoost model. Run:

- Permutation importance
- TreeSHAP global
- PDP for top-3 features
- One LIME explanation

Note where they agree and disagree.

2.11 Quiz

1. Place SHAP for one prediction on all four axes.
2. Why is Grad-CAM model-specific?
3. Define faithfulness in one sentence.
4. Why does the disagreement problem matter for compliance?
5. Give two examples of intrinsic-interpretable models that can match black-box accuracy on tabular tasks.

Unit 2

Feature Attribution and Gradient-Based Methods

Required Contact Hours: **8**

- LIME — local linear surrogate via perturbation
- SHAP — Shapley values, KernelSHAP, TreeSHAP, DeepSHAP
- Anchors and Influence functions
- Gradient attribution — saliency, SmoothGrad, Integrated Gradients
- Grad-CAM, Grad-CAM++, Score-CAM

The chapters that follow expand each topic above with theory, worked examples, reading lists, hands-on lab guidance, and end-of-chapter self-assessment questions.

XAI Module 3 — LIME, SHAP, and the Feature Attribution Family

> ***Goal**: Master the two most-cited XAI methods. Be able to implement both from scratch in ~100 lines of Python each.*

3.1 LIME — Local Interpretable Model-agnostic Explanations

Ribeiro, Singh, Guestrin — KDD 2016. **The first widely-adopted post-hoc method.**

Idea

For a single instance `x`, fit a simple, interpretable surrogate model `g` (e.g., sparse linear) in the local neighborhood of `x` such that `g` approximates `f` there. The surrogate's coefficients become the explanation.

Algorithm

```
Input: model f, instance x, num_samples N
1. Sample N perturbations x'_i in the neighborhood of x.
   - Tabular: perturb features around x.
   - Text: randomly delete words.
   - Image: superpixels (SLIC) toggled on/off.
2. Compute model predictions y'_i = f(x'_i).
3. Weight each sample by a kernel π(x, x'_i) — closer to x → higher weight.
4. Fit a linear model g on (x'_i, y'_i) weighted by π and with sparsity (L1).
5. Return g's coefficients as the explanation.
```

Strengths

- Truly model-agnostic.
- Intuitive output (sparse linear).

Weaknesses

- **Instability** — different random seeds give different explanations (Alvarez-Melis & Jaakkola, 2018).
- **Choice of neighborhood** is arbitrary and influences results.
- **Adversarial vulnerability** — Slack et al. (2020) showed a model can be made to lie to LIME while still being biased.

3.2 SHAP — SHapley Additive exPlanations

Lundberg & Lee — NeurIPS 2017. **The single most-cited XAI method**.

Idea

Borrow Shapley values from cooperative game theory. For each feature, ask: "What is its marginal contribution to the prediction, averaged over all possible coalitions of other features?"

Shapley value formula

For player i in game v with N players:

```
φ_i(v) = Σ_{S ⊆ N \ {i}}  [ |S|! · (|N|-|S|-1)! / |N|! ] · [ v(S ∪ {i}) - v(S) ]
```

In ML: `v(S) = E[f(X) | X_S = x_S]` — expected prediction conditioned on the features in S being fixed at x's values.

Why Shapley values matter

They are the **unique** allocation satisfying:

1. **Efficiency** — $\Sigma\, \varphi_i = f(x) - E[f(X)]$ (attributions sum to the gap from baseline)
2. **Symmetry** — symmetric features get equal credit
3. **Dummy** — irrelevant features get 0
4. **Linearity** — for two games combined, attributions add

This axiomatic uniqueness is what gives SHAP its theoretical edge over LIME.

Variants

Variant	Speed	Faithfulness
KernelSHAP	Slow (model-agnostic via weighted regression)	Good
TreeSHAP	Fast (exact, exploits tree structure)	Exact for tree ensembles
DeepSHAP / Deep LIFT	Medium (chain rule with reference)	Good for NNs
PartitionSHAP	Medium (hierarchical Owen values)	Useful for high-dim features
FastSHAP	Very fast (amortized via a learned predictor)	Good after training

Conditional vs marginal expectation

- **Marginal** (`E[f | do(X_S = x_S)]`) — break dependence, like in interventional causal inference.
- **Conditional** (`E[f | X_S = x_S]`) — preserve dependence.

Aas et al. (2021) showed marginal SHAP can credit features that aren't in the model. Use conditional when features are correlated. Most libraries default to marginal — beware.

3.3 SHAP visualizations to know

- **Force plot** — per-prediction, additive.
- **Waterfall plot** — per-prediction, sequential.
- **Summary / beeswarm plot** — global, distribution of attributions across data.
- **Dependence plot** — feature value vs SHAP value with color = interacting feature.
- **Decision plot** — multi-prediction comparison.

3.4 LIME vs SHAP — when to use which

Use case	Pick
Tree ensemble (XGBoost, LightGBM, RF)	**TreeSHAP** (fast, exact)
Black box, prototype phase	LIME (faster setup)
Need theoretical guarantees	SHAP
Need to explain text or images	Either, but consider gradient methods too
Tight compute budget	LIME
Compliance documentation	SHAP (axiomatic story is easier to defend)

3.5 Anchors — Ribeiro et al. 2018

A rule-based local explanation: "Anchor" = a set of feature predicates such that, whenever those predicates hold, the model's prediction is the same with high probability (e.g., precision $\geq$ 0.95).

```
IF age > 50 AND education = "Bachelor"  THEN predict = "income > 50K"  (precision 0.97,
```

Pros: high-precision, easy to read. Cons: anchors may not exist for all instances; coverage trade-off.

3.6 DeepLIFT / Layer-wise Relevance Propagation (LRP)

Both methods backpropagate attributions through a neural network using a reference / baseline input, distributing credit to inputs.

- **DeepLIFT** (Shrikumar et al., 2017) — chain rule with multipliers relative to reference.
- **LRP** (Bach et al., 2015) — propagate relevance backward via custom rules per layer.

Both are precursors to Integrated Gradients and underlie DeepSHAP.

3.7 Influence functions — Koh & Liang (2017)

A different kind of explanation: *which training points most influenced this prediction?*

Approximates leave-one-out retraining without retraining, via implicit Hessian-vector products.

Useful for debugging mislabeled data, finding adversarial training examples, and credit attribution. Computationally expensive for large models; scalable variants (TracIn, Datamodels) are active research.

3.8 Implementing KernelSHAP in 60 lines (sketch)

```
import numpy as np
from itertools import combinations
from math import comb

def kernel_weight(M, s):
    if s == 0 or s == M: return 1e6   # large weight, edge case
    return (M - 1) / (comb(M, s) * s * (M - s))

def kernel_shap(f, x, background, num_samples=2000):
    M = len(x)
    masks, weights, ys = [], [], []
    for _ in range(num_samples):
        z = np.random.randint(0, 2, M)
        x_masked = np.where(z, x, background.mean(0))
        masks.append(z)
        weights.append(kernel_weight(M, z.sum()))
        ys.append(f(x_masked))
    Z = np.array(masks); w = np.array(weights); y = np.array(ys)
    # Weighted linear regression with intercept = E[f]
    base = f(background).mean()
    y_centered = y - base
    W = np.diag(w)
    phi = np.linalg.lstsq(Z.T @ W @ Z, Z.T @ W @ y_centered, rcond=None)[0]
    return phi  # one Shapley value per feature
```

This is enough to convince yourself the math works. Production SHAP is much more optimized.

3.9 Reading list

- Ribeiro, Singh, Guestrin — *"Why Should I Trust You?": Explaining the Predictions of Any Classifier*, KDD 2016.
- Lundberg & Lee — *A Unified Approach to Interpreting Model Predictions*, NeurIPS 2017.
- Lundberg et al. — *From local explanations to global understanding with explainable AI for trees*, Nature MI 2020.
- Ribeiro, Singh, Guestrin — *Anchors: High-Precision Model-Agnostic Explanations*, AAAI 2018.
- Shrikumar, Greenside, Kundaje — *Learning Important Features Through Propagating Activation Differences (DeepLIFT)*, ICML 2017.
- Aas, Jullum, Løland — *Explaining individual predictions when features are dependent*, 2021.
- Slack et al. — *Fooling LIME and SHAP*, AIES 2020.
- Koh & Liang — *Understanding Black-box Predictions via Influence Functions*, ICML 2017.

3.10 Lab (3 hrs)

1. Implement KernelSHAP from scratch (use the sketch above as starter).
2. Verify it matches `shap` library output on 5 examples from a small XGBoost model.
3. Run TreeSHAP on the same model; compare runtime.
4. Implement an "adversarial" model that detects when it's being explained and returns a different (benign) function (per Slack et al.).

3.11 Quiz

1. State the four Shapley axioms.
2. Why is TreeSHAP exact and KernelSHAP only approximate?
3. Explain marginal vs conditional SHAP and when each is appropriate.
4. How does LIME's neighborhood choice affect explanations?
5. What is an "anchor" and when does it not exist?

XAI Module 4 — Gradient-Based Attribution

> ***Goal***: *Master the saliency / gradient family — the workhorse of computer-vision XAI.*

4.1 The core intuition

If you change pixel `x_i` by a tiny amount, how much does the output `f(x)` change?

That's the partial derivative `∂f/∂x_i`. Gradient-based methods are variations on this single idea, fixing its (many) flaws.

4.2 Vanilla saliency (Simonyan et al., 2013)

```
saliency(x_i) = | ∂f_c(x) / ∂x_i |
```

Visualize as a heatmap over the input image. Simplest possible attribution.

Problems

- Noisy — gradients fluctuate wildly across nearby inputs.
- Saturation — for ReLU networks, large activations zero out the gradient even when the feature matters.
- Doesn't distinguish *what* makes the prediction (positive vs negative evidence).

4.3 SmoothGrad (Smilkov et al., 2017)

Average saliency over noisy copies:

```
SmoothGrad(x) = (1/N) · Σ saliency(x + ε_i), ε_i ~ N(0, σ²)
```

Visually much cleaner. Essentially a denoising of vanilla saliency.

4.4 Integrated Gradients (Sundararajan, Taly, Yan — ICML 2017)

Solve the saturation problem by integrating along a straight path from a baseline `x'` to input `x`:

```
IG_i(x) = (x_i − x'_i) · ∫_{α=0}^{1}  ∂f(x' + α(x − x')) / ∂x_i  dα
```

Approximated by Riemann sum with ~50 steps.

Why IG matters

Satisfies two axioms:

1. **Sensitivity** — if changing a feature changes prediction, attribution is nonzero.
2. **Implementation invariance** — two functionally-equivalent networks yield identical attributions.

Vanilla saliency violates Sensitivity (saturation). Pre-IG methods (DeConvNet, Guided Backprop) violate Implementation Invariance.

Choice of baseline matters

Baseline	Use case
All-zeros image	Default but problematic for "black" matters
Random noise	Hides biases
Blurred version of x	"What changed from blurry-known to detailed-classified"
Mean of training set	Centroidal
Multiple baselines, averaged	Most robust (Expected Gradients)

Expected Gradients (Erion et al., 2021) averages over a distribution of baselines — combines IG with SHAP's expectation.

4.5 Guided Backprop & DeConvNet

Modify the backward pass through ReLU to suppress negative gradients. Visually nice but fails sanity checks (Adebayo et al., 2018) — they produce similar-looking saliency maps **even when the model is randomly initialized**.

Conclusion: Guided BP is essentially an edge detector, not a faithful explanation.

4.6 Grad-CAM (Selvaraju et al., ICCV 2017)

Compute the gradient of the target class score with respect to the activations of the **last conv layer**:

```
α_k = (1/Z) · Σ_{i,j} ∂y_c / ∂A^k_{i,j}
Grad-CAM = ReLU(Σ_k α_k · A^k)
```

Produces a low-resolution (e.g., 7×7) heatmap upsampled to the input. **The most-used saliency method in practice**.

Variants

- **Grad-CAM++** — better localization when multiple instances of the class are present.
- **Score-CAM** — replaces gradients with forward-pass weights (more faithful).
- **Eigen-CAM** — uses the first principal component of activation maps.
- **HiResCAM** — corrects a small flaw in Grad-CAM's element-wise multiplication.

4.7 Layer-wise Relevance Propagation (LRP)

A backward propagation rule for redistributing the prediction relevance through the network. Different "rules" (LRP-ε, LRP-γ, LRP-αβ) work best for different layer types.

Most popular for clinical/medical imaging because it's deterministic and faithful by construction (relevance conservation across layers).

4.8 Sanity checks (Adebayo et al., NeurIPS 2018)

A paper everyone should read once.

Two tests:

1. **Model randomization test** — re-initialize weights from the top layer down. A faithful explanation should change drastically.
2. **Data randomization test** — train on permuted labels. A faithful explanation should produce different attributions than the correctly-trained model.

Result: Guided Backprop and Guided Grad-CAM **fail** both tests. Integrated Gradients and vanilla saliency pass model-randomization but only partially data-randomization.

Takeaway: a pretty heatmap may be edge-detection, not explanation.

4.9 Attribution for Transformers

Gradient methods generalize to transformers but care is needed:

- Saliency works token-wise.
- Integrated Gradients with token embedding baselines.
- **Attention Rollout** (Abnar & Zuidema 2020) — multiply attention matrices across layers.
- **Attention Flow** — max-flow through attention graph.
- **Chefer et al.** (2021) — generic Transformer attribution combining gradients + attention.

(More in Module 8.)

4.10 The IG family today

The 2024–2026 SOTA cluster:

- **Path-Integrated Gradients with optimal paths**
- **Bias-Free Integrated Gradients**
- **GAM (Guided Attention Maps)** for ViTs
- **CAFE / TracIn-Influence** for input + training attribution

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4.11 Reading list

- Simonyan, Vedaldi, Zisserman — *Deep Inside Convolutional Networks: Visualising Image Classification Models*, 2013.

- Sundararajan, Taly, Yan — *Axiomatic Attribution for Deep Networks (Integrated Gradients)*, ICML 2017.
- Smilkov et al. — *SmoothGrad: removing noise by adding noise*, 2017.
- Selvaraju et al. — *Grad-CAM*, ICCV 2017.
- Bach et al. — *On Pixel-Wise Explanations for Non-Linear Classifier Decisions by Layer-Wise Relevance Propagation*, PLoS ONE 2015.
- Adebayo et al. — *Sanity Checks for Saliency Maps*, NeurIPS 2018.
- Erion et al. — *Improving performance of deep learning models with axiomatic attribution priors and expected gradients*, Nature MI 2021.

4.12 Lab (3 hrs)

1. Use a pretrained ResNet-50 on an ImageNet image.
2. Implement and visualize: vanilla saliency, SmoothGrad, Integrated Gradients (50 steps), Grad-CAM.
3. Run Adebayo's randomization test on each method. Report which survive.
4. Repeat the IG step with three different baselines. Discuss differences.

4.13 Quiz

1. What does the "implementation invariance" axiom rule out?
2. Why does Grad-CAM produce a low-resolution map?
3. State the SmoothGrad equation.
4. Explain in one sentence why Guided Backprop fails the sanity check.
5. Why is baseline choice central to IG's interpretation?

Unit 3

Concept-Based, Counterfactual and Global Methods

Required Contact Hours: **7**

• TCAV, ACE, CRAFT, Network Dissection

• Concept Bottleneck Models, ProtoPNet

• Counterfactual explanations — Wachter, DiCE, FACE

• Algorithmic recourse and contrastive explanations

• Global methods — PDP, ICE, ALE, surrogates, ANOVA

The chapters that follow expand each topic above with theory, worked examples, reading lists, hands-on lab guidance, and end-of-chapter self-assessment questions.

XAI Module 5 — Concept-Based Explanations

> ***Goal**: Move from pixel-level attribution to **human-meaningful concepts** (e.g., "stripes", "wheel", "femininity in a face").*

5.1 The motivation

Pixel-level saliency tells you *where*, not *what*. A doctor wants "the model flagged this because of irregular cell shape", not "pixel (243, 187) was important".

Concept-based methods answer **what concept drove the prediction**.

5.2 TCAV — Testing with Concept Activation Vectors

Kim, Wattenberg, Gilmer, Cai, Wexler, Viegas, Sayres — ICML 2018.

Idea

1. **Define a concept** by collecting a set of positive examples (e.g., 50 striped images) and random negatives.
2. **Compute activations** at a chosen layer `l` for both sets.
3. **Train a linear classifier** to separate concept vs random in activation space → the normal to this hyperplane is the **Concept Activation Vector (CAV)**.
4. **Compute directional derivatives** of the target class prediction with respect to the CAV.
5. **TCAV score** = fraction of input samples for which the directional derivative is positive.

```
S_{C,k,l}(x) = ∂h_{l,k}(f_l(x)) / ∂C        (directional derivative)
TCAV_{C,k,l} = | {x in X_k : S_{C,k,l}(x) > 0} | / | X_k |
```

A statistical significance test (vs random concepts) rules out spurious results.

Why TCAV matters

- Uses human-understandable concepts, not pixels.
- Works for any pretrained model.
- Established a research line that includes ACE, CRAFT, Network Dissection, Concept Bottleneck Models.

Limits

- Requires curated concept datasets.
- Concept linear-separability assumption.
- Can suffer from concept entanglement.

5.3 ACE — Automated Concept Extraction

Ghorbani, Wexler, Zou, Kim (NeurIPS 2019).

Automates the "where do concepts come from" step:

1. Segment images into superpixels.
2. Cluster segments in activation space.
3. Treat each cluster centroid as an auto-discovered concept.
4. Run TCAV against it.

ACE finds concepts the researcher might not have thought to define.

5.4 CRAFT — Fel et al. (CVPR 2023)

Combines:

- Recursive non-negative matrix factorization (NMF) of activations
- Per-region attribution
- Per-region concept attribution

Output: a hierarchical decomposition of the prediction by concept, with localization. State of the art for concept-based CV explanations as of 2024.

5.5 Network Dissection (Bau, Zhou, Khosla, Oliva, Torralba — CVPR 2017)

For each conv filter, find the human-labeled concept (from the Broden dataset: objects, parts, materials, textures, scenes) it best aligns with. Result: assign filters to concepts.

Shows that some networks have surprisingly "monosemantic" units even without being trained for it.

Inspired modern **mechanistic interpretability** (see Module 9).

5.6 Concept Bottleneck Models (Koh et al., ICML 2020)

Train the model in two stages:

```
x → C(x) = predicted concepts → ŷ
```

The concept layer is **explicit** — humans can inspect and even *intervene* on it ("test-time concept editing"). If the model says "this bird has white belly + curved beak" and you know that's wrong, you can correct it before the final prediction.

Trade-off: requires concept labels for training; accuracy gap if concepts are insufficient.

Post-hoc CBMs (Yuksekgonul, Wang, Zou — NeurIPS 2022) — convert any trained model into a CBM by fitting a concept layer on its features. Much cheaper.

5.7 Prototype networks — "this looks like that"

Chen et al. — *This Looks Like That: Deep Learning for Interpretable Image Recognition*, NeurIPS 2019.

ProtoPNet: the network learns a set of prototypes per class. Predictions are explained as "this part of input looks like this prototype of training class X".

Visually beautiful, used in medical imaging. Variants: ProtoTree, Deformable ProtoPNet, ProtoPool.

5.8 Concept Bottleneck for LLMs

Active 2024–2026 area:

- **Concept embeddings** for transformer features.
- **Sparse Autoencoders (SAEs)** decompose activations into thousands of monosemantic concepts (Anthropic, Bricken et al. 2023; Cunningham et al. 2024; Templeton et al. 2024 — *Scaling monosemanticity*).
- Probing classifiers find what concepts are encoded at each layer.

This is the most active research thread in XAI right now — see Module 9.

5.9 Limits & critiques

- **Concept arbitrariness** — different researchers pick different concept sets.
- **Concept leakage** — concept labels may encode unintended info (e.g., demographics).
- **Faithfulness** — concept predictors may not reflect what the underlying model actually uses (Margeloiu et al., 2021).
- **Concept entanglement** — single filters often respond to multiple concepts.

5.10 Reading list

- Kim et al. — *Interpretability Beyond Feature Attribution: Quantitative Testing with CAVs (TCAV)*, ICML 2018.
- Ghorbani et al. — *Towards Automatic Concept-based Explanations (ACE)*, NeurIPS 2019.
- Fel et al. — *CRAFT: Concept Recursive Activation FacTorization*, CVPR 2023.
- Bau et al. — *Network Dissection*, CVPR 2017.
- Koh et al. — *Concept Bottleneck Models*, ICML 2020.
- Chen et al. — *This Looks Like That*, NeurIPS 2019.
- Bricken et al. — *Towards Monosemanticity*, Anthropic 2023.
- Templeton et al. — *Scaling Monosemanticity*, Anthropic 2024.

5.11 Lab (3 hrs)

1. Pick a small ImageNet subset.
2. Build a "stripes" concept set (50 images of stripes) and a random negative set.
3. Compute the CAV at the last conv layer of ResNet-50.
4. Run TCAV for class "zebra" — verify that "stripes" matters; check significance vs random concepts.
5. Repeat for a class where "stripes" should *not* matter (e.g., "spaniel").

5.12 Quiz

1. Why is TCAV more useful than saliency for clinical communication?
2. What does the TCAV score quantify and how is significance tested?
3. State two differences between standard CBMs and post-hoc CBMs.
4. How does ProtoPNet explain a prediction?
5. What does the "monosemanticity" research line aim to demonstrate?

XAI Module 6 — Counterfactual & Contrastive Explanations

> ***Goal***: *Master the "what-if" family of explanations — arguably the most actionable form of XAI.*

6.1 The contrastive idea

People naturally explain by contrast: "I was denied the loan because my income was too low" implicitly says "if my income had been $X higher, I would have been approved".

Counterfactual explanations operationalize this. They are:

- **Actionable** ("change feature X to Y")
- **Aligned with how humans actually explain** (Miller, 2019)
- **Stakeholder-friendly** (no model internals required)

6.2 Wachter, Mittelstadt, Russell (2017) — the canonical formulation

> *"What is the smallest change to `x` that would change the prediction to `y'`?"*

```
x_cf = argmin_{x'}  d(x, x') + λ · L(f(x'), y')
```

Where:

- `d(x, x')` = distance (often weighted L1 on normalized features)
- `L(f(x'), y')` = loss pushing prediction to desired class

Optimized via gradient descent for differentiable models or genetic/discrete search for non-differentiable ones.

Desiderata

A good counterfactual is:

1. **Valid** — actually flips the prediction.
2. **Proximate** — close to `x` in input space.
3. **Sparse** — changes few features.
4. **Actionable** — only changes mutable features (can't change race, age).
5. **Plausible** — lies on the data manifold (don't recommend "earn $1,000,000 next month").
6. **Diverse** — multiple distinct paths to flip the decision.

6.3 DiCE — Diverse Counterfactual Explanations

Mothilal, Sharma, Tan — FAT* 2020.

Generates K counterfactuals that are simultaneously:

- Valid
- Close to `x`
- Mutually **diverse** (via DPP / determinantal-point-process-like regularizer)
- Respect feature constraints (mutable / immutable, monotonic)

Library: `dice-ml`. Widely used in finance / lending.

6.4 FACE — Feasible and Actionable Counterfactual Explanations

Poyiadzi et al. — AIES 2020.

Builds a graph over training data; the counterfactual is the shortest path from `x` to a flipping region through high-density regions. Ensures **plausibility** by construction.

6.5 GeCo — Genetic Counterfactuals

Searches counterfactual space via genetic algorithm with mutation, crossover, and constraint handling. Works for arbitrary black-box models including non-differentiable trees.

6.6 MACE — Model-Agnostic Counterfactual Explanations via SAT/SMT

Karimi et al. — FAT* 2020.

Encode the model + counterfactual desiderata as a SAT/SMT problem. Provably optimal counterfactuals for piecewise-linear models (trees, ReLU nets).

6.7 Algorithmic recourse — the causal turn

Karimi, Schölkopf, Valera (FAccT 2021) point out: classical counterfactuals ignore causal structure. "Change your education from BS to PhD" isn't a single intervention — it requires years and money.

Recourse = the **actions** required to achieve the counterfactual, accounting for causal effects.

```
intervention I  →  features x' (downstream causal effects propagate)  →  prediction f(x'
```

Requires a causal graph (often unavailable in practice — see "recourse under uncertain causal knowledge").

6.8 Contrastive explanations (pertinent positives / negatives)

Dhurandhar et al. — NeurIPS 2018, "Explanations based on the Missing".

Two halves of any prediction:

- **Pertinent positives (PP)** — minimal subset of features that *sufficiently* causes the prediction.
- **Pertinent negatives (PN)** — features whose absence is necessary (their addition would flip).

A complete contrastive explanation: "Prediction is X because of PP, and not Y because of PN."

6.9 Counterfactuals for NLP

Method	Approach
Polyjuice (Wu et al. 2021)	GPT-fine-tuned counterfactual generator
CheckList (Ribeiro et al. 2020)	Templated minimal pairs for behavioral testing
MiCE (Ross et al. 2021)	Minimal contrastive edits via Levenshtein
CAT	LLM-as-editor for counterfactual NLP

Used to find spurious correlations (e.g., model predicts "positive" for *any* sentence with the word "amazing").

6.10 Counterfactuals for images

Hard problem — pixel space is high-dimensional and "small change" is meaningless without a manifold prior.

Approach	Method
Latent-space (StyleGAN inversion)	Edit w vector → decode
Diffusion-based	Counterfactual via guided diffusion
Saliency-region edits	Inpaint salient region differently
Concept-aware (e.g., StylEx, Lang et al. 2021)	GAN with classifier-aligned latent directions

---

6.11 Counterfactuals as compliance tools

Wachter et al. argued counterfactuals satisfy GDPR's right-to-explanation. They don't reveal model internals (good for trade secrets) but give the user actionable information.

Adopted in:

- Credit denial letters (US ECOA adverse-action notices)
- Algorithmic hiring tools
- Insurance underwriting

6.12 Failures and critiques

- **Manifold off-distribution** — gradient-descent counterfactuals often produce inputs no human would recognize.
- **Adversarial-like artifacts** — minimum-change counterfactuals can be adversarial perturbations.
- **Causal naivety** — ignoring causal relations gives unactionable advice.
- **Fairness** — counterfactual recommendations can be discriminatory if mutable features differ across groups

(e.g., harder for some groups to "increase income").

- **Multiplicity** — many equally-good counterfactuals exist; which do you show?

6.13 Reading list

- Wachter, Mittelstadt, Russell — *Counterfactual Explanations Without Opening the Black Box*, Harvard JOLT 2017.
- Mothilal, Sharma, Tan — *Explaining Machine Learning Classifiers through Diverse Counterfactual Explanations (DiCE)*, FAT* 2020.
- Poyiadzi et al. — *FACE: Feasible and Actionable Counterfactual Explanations*, AIES 2020.
- Karimi et al. — *A Survey of Algorithmic Recourse*, ACM Computing Surveys 2022.
- Dhurandhar et al. — *Explanations based on the Missing*, NeurIPS 2018.
- Ross, Marasović, Peters — *Explaining NLP Models via Minimal Contrastive Editing*, ACL 2021.
- Verma, Dickerson, Hines — *Counterfactual Explanations and Algorithmic Recourse for Machine Learning: A Review*, 2024.

6.14 Lab (3 hrs)

1. Train a logistic regression on UCI Adult (income $\geq$ \$50k).
2. For 10 denied applicants, generate counterfactuals using:
    - Vanilla Wachter (gradient)
    - DiCE
3. Audit: which features did each method change? Are they mutable / actionable?
4. Discuss fairness — do counterfactuals for women differ systematically from men's?

6.15 Quiz

1. State Wachter's objective in math.
2. Distinguish counterfactual explanation from algorithmic recourse.
3. Why is plausibility hard to enforce via gradient descent alone?
4. Define pertinent positives and pertinent negatives.
5. Why might counterfactuals discriminate even when the model itself is "fair"?

XAI Module 7 — Global Methods: PDP, ICE, ALE, Surrogates, Permutation Importance

> ***Goal***: *Master the global / cohort-level methods for tabular ML — the bread and butter of every applied XAI engagement.*

7.1 Partial Dependence Plots (PDP)

Friedman, 2001.

For feature(s) `S`, marginalize out the rest:

```
PDP_S(x_S) = (1/N) Σ_i f(x_S, x_i^{(C)})
```

For each value of `x_S`, average the prediction over the dataset, holding `x_S` fixed.

Visualization

- 1D feature → line plot.
- 2D pair → heatmap (shows interactions).

Limits

- **Assumes feature independence** — if `x_S` is correlated with `x_C`, the marginalization creates impossible synthetic data points (a 6-foot 60-pound person).
- Only shows **average** effect — hides heterogeneity.

7.2 ICE — Individual Conditional Expectation

Goldstein et al., 2015.

Instead of averaging, plot **one line per instance**: how does *this person's* prediction change as we vary `x_S` for them?

PDP = average of all ICE curves.

When ICE curves diverge significantly, the PDP is misleading and you have heterogeneous effects.

7.3 ALE — Accumulated Local Effects

Apley & Zhu, 2020.

Solves PDP's correlation problem.

```
For interval k of feature x_j:
   compute local effect = E[ f(x_j = upper) - f(x_j = lower) | x_j in interval k ]
```

```
Accumulate and center across intervals.
```

ALE conditions on the actual data distribution, so it doesn't fabricate impossible samples. **Should be the default for correlated features.**

Method	Handles correlation?	Computational cost
PDP	No	Low
ICE	No	Low
ALE	Yes	Low
SHAP global	Conditionally yes	Medium

7.4 Permutation Feature Importance

Breiman, 2001 (random forests); model-agnostic since.

```
Importance(j) = error(f, X_permuted_j, y) - error(f, X, y)
```

Shuffle one feature's column, measure performance drop. Bigger drop = more important feature.

Pitfalls

- Same correlation issue (creates impossible data points).
- Importance is **with respect to the loss**, not the prediction — interpret accordingly.
- High-cardinality features can be over-credited (use grouped permutation).

Conditional permutation importance fixes some of these by shuffling within strata.

7.5 Global SHAP

Aggregate local SHAP values across the dataset:

- Mean(|SHAP|) per feature → global importance
- Summary / beeswarm plot → distribution
- Dependence plots → effect across the data

This is "the SHAP plot" you see in 90% of Kaggle kernels.

7.6 Surrogate models

Train a simpler, interpretable model to mimic the black box.

Global surrogate

```
f_black_box(x) → ŷ
g_interpretable(x) → ŷ  (regress g on f's predictions over a large sample)
```

If `g`'s $R^2 \approx 1$, you can use `g` as the explanation. Otherwise, `g`'s explanations are unreliable.

Knowledge distillation

A specific form of global surrogate where a *smaller neural net* mimics a larger one. Hinton et al., 2015. Not strictly XAI but related.

Soft Decision Trees (Frosst & Hinton, 2017)

Trees with stochastic routing that can match a neural net while remaining traceable.

7.7 Functional ANOVA decomposition

Decompose `f(x) = f_∅ + Σ f_i(x_i) + Σ f_{ij}(x_i, x_j) + ...` into main effects + interactions.

EBMs do this exactly — and that's why they tend to be both accurate and interpretable on tabular data.

7.8 H-statistic for interactions

Friedman & Popescu, 2008. Measures whether feature pair `(j, k)` interacts beyond their additive effects.

```
H²_jk = Σ_i [ PDP_jk(x_i) − PDP_j(x_i) − PDP_k(x_i) ]² / Σ_i PDP_jk(x_i)²
```

Useful when you suspect interactions are driving the model.

7.9 Anchors at the global level

Aggregate Anchors across the dataset to surface high-precision rule patterns the model uses.

7.10 Functional cohorts — SliceFinder

Chung, Polyzotis, Tae, Whang (2019) — automatically find data slices where the model performs worst. Critical for fairness audits.

```
slice = {x : age > 60 AND device = "android"}
   if loss on slice >> overall loss → investigate
```

Tools: Slice Finder, Aequitas, FairLearn, Robustness Gym.

7.11 When to use which

Task	Use
Quick global feature rank	Permutation importance or global SHAP
Visualize a single feature's effect	PDP + ICE (small dataset), ALE (large/correlated)
Inspect interaction	PDP-2D or H-statistic
Build an explanation for stakeholders	Surrogate tree
Audit subgroup performance	SliceFinder + Aequitas
Need axiomatic guarantees	SHAP global

7.12 Reading list

- Friedman — *Greedy Function Approximation: A Gradient Boosting Machine*, Annals 2001 (PDP section).
- Goldstein et al. — *Peeking Inside the Black Box: Visualizing Statistical Learning with ICE*, JCGS 2015.
- Apley & Zhu — *Visualizing the Effects of Predictor Variables in Black Box Supervised Learning Models (ALE)*, JRSS B 2020.
- Breiman — *Random Forests*, ML 2001 (permutation importance).
- Friedman & Popescu — *Predictive Learning via Rule Ensembles*, AOAS 2008.
- Chung et al. — *Slice Finder: Automated Data Slicing for Model Validation*, ICDE 2019.
- Molnar — *Interpretable Machine Learning*, Ch 5–7.

7.13 Lab (2 hrs)

1. On the California Housing dataset, train XGBoost.
2. Generate PDP, ICE, ALE, permutation importance, global SHAP plots for the top-5 features.
3. Find at least one feature where PDP and ALE disagree → explain why (correlation).
4. Use SliceFinder to find the worst-performing data slice.

7.14 Quiz

1. Why can PDP be misleading for correlated features?
2. State the formal difference between PDP and ICE.
3. When would you choose ALE over SHAP?
4. Give one situation where permutation importance over-credits a feature.
5. What's the relationship between EBMs and functional ANOVA?

Unit 4

Attention, Transformer and Mechanistic Interpretability

Required Contact Hours: **9**

- Attention Is (Not) Explanation debate
- Attention Rollout, Flow, Chefer et al.
- Probing classifiers, Logit Lens, Tuned Lens
- Distill Circuits, Mathematical Framework for Transformer Circuits
- Induction heads, IOI circuit, ROME knowledge editing
- Sparse Autoencoders, monosemanticity, ACDC

The chapters that follow expand each topic above with theory, worked examples, reading lists, hands-on lab guidance, and end-of-chapter self-assessment questions.

XAI Module 8 — Attention, Transformer & LLM Interpretability

> ***Goal***: *Understand the methods for interpreting transformer-based models — the dominant architecture of the LLM era.*

8.1 The attention-as-explanation debate

In 2019, two papers fought:

- **Jain & Wallace** — *Attention Is Not Explanation* (NAACL 2019). Shows attention weights can be radically altered without changing predictions.
- **Wiegreffe & Pinter** — *Attention Is Not Not Explanation* (EMNLP 2019). Shows attention is often informative, even if not the unique explanation.

Consensus (2021–2026):

- Raw attention weights are *not* faithful explanations on their own.
- Combined with gradients (Grad×Attention, Chefer et al. 2021), they can be faithful.
- Better: use methods designed for transformers (Attention Rollout, gradient-weighted attention).

8.2 Attention Rollout (Abnar & Zuidema, 2020)

Multiply attention matrices across layers, adding the identity for residual flow:

```
A_l_rollout = (A_l + I) · A_{l-1}_rollout
normalize each row
```

Yields a single matrix showing each token's accumulated influence on the [CLS] / output token. Used heavily for ViT interpretation.

8.3 Attention Flow

Treat the network as a graph where edges = attention weights. Compute max-flow from input tokens to output. Slower than rollout but cleaner attribution.

8.4 Generic Transformer Attribution (Chefer et al., CVPR 2021)

A method-of-choice for ViTs combining:

- Gradients with respect to attention
- Layer-wise relevance propagation rules
- Aggregation across heads and layers

Produces clean, faithful heatmaps for both encoder-only (BERT/ViT) and encoder-decoder transformers.

8.5 Probing classifiers

Train a simple classifier (logistic regression) on top of intermediate layer activations to predict some property `P` (POS tag, syntactic role, sentiment).

If the probe scores high → the model encodes `P` at that layer.

Pitfalls

- Probe accuracy $\neq$ model use of the property.
- Hewitt & Liang (2019) — *control task* baselines necessary to separate "info is there" from "info is usable".

8.6 Logit Lens (nostalgebraist, 2020 → formalized later)

Apply the unembedding matrix to intermediate residual stream activations to see what tokens the model "would predict" at each layer.

Shows the gradual emergence of the prediction — useful for hallucination analysis.

Tuned Lens (Belrose et al., 2023)

Train a small learned linear probe per layer instead of using the unembedding directly. Smoother trajectories, less noise.

8.7 Activation patching / causal tracing

ROME (Meng et al., NeurIPS 2022). To find *where* in the network a fact lives:

1. Run a forward pass with the original input → cache activations.
2. Run with corrupted input (e.g., "Eiffel Tower" → "Statue of Liberty").
3. Patch in cached clean activations at one site at a time.
4. Site that recovers correct answer → "this is where the fact lives".

Why this matters

Localized facts can then be **edited** (ROME, MEMIT, GRACE) — opening model editing as a discipline. Hot 2023–2026.

8.8 The induction-head story (Anthropic, 2022)

Olsson et al., *In-context Learning and Induction Heads*. Identified two-layer attention heads that implement [A][B] ... [A] → [B] copying — the basic mechanism of in-context learning.

This is the **canonical mechanistic-interpretability case study**. Read it.

8.9 Sparse Autoencoders (SAEs) — the 2023–2025 breakthrough

The residual stream of a transformer carries many superimposed features (superposition hypothesis, Elhage et al. 2022). SAEs decompose activations into **sparse, monosemantic features**.

Pipeline:

1. Pick a layer; collect millions of activations.
2. Train an SAE: `encoder` to k-sparse latent, `decoder` to reconstruct.
3. Each learned feature → one neuron in latent space — often human-meaningful.
4. Use **feature attribution** and **steering** to test causality.

Papers:

- Bricken et al. (Anthropic) — *Towards Monosemanticity*, 2023.
- Cunningham et al. — *Sparse Autoencoders Find Highly Interpretable Features in LMs*, ICLR 2024.
- Templeton et al. (Anthropic) — *Scaling Monosemanticity to Claude 3 Sonnet*, 2024.

This is the **best lead** the field has on understanding LLMs as of 2026.

8.10 LLM-specific interpretability tools

Tool	Purpose
TransformerLens	Hooks & interventions for HF models
nnsight	Remote interventions on large models (NDIF)
SAELens / Neuronpedia	Browse SAE features
Garcon / Inspect (Anthropic)	Internal Anthropic tooling, partly OSS
bertviz	Attention head visualization
OpenAI Microscope (legacy)	Neuron visualizations for vision models

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8.11 Hallucination explanation

Open problem: *why* does a given LLM hallucinate?

Approaches:

- Self-consistency (multiple samples → disagreement = uncertainty)
- Logit Lens for divergence between layers
- SAE feature attribution at the hallucination token
- RAG attribution (which retrieved chunks influenced the output?)

Industry deployment (Anthropic Claude Citations, OpenAI Structured Outputs) now ships citation-grounded outputs as a partial answer.

8.12 Reading list

- Vaswani et al. — *Attention Is All You Need*, NeurIPS 2017 (architecture baseline).
- Jain & Wallace — *Attention Is Not Explanation*, NAACL 2019.
- Wiegreffe & Pinter — *Attention Is Not Not Explanation*, EMNLP 2019.
- Abnar & Zuidema — *Quantifying Attention Flow in Transformers*, ACL 2020.
- Chefer, Gur, Wolf — *Transformer Interpretability Beyond Attention Visualization*, CVPR 2021.
- Meng et al. — *Locating and Editing Factual Associations in GPT (ROME)*, NeurIPS 2022.
- Elhage et al. — *A Mathematical Framework for Transformer Circuits*, Anthropic 2021.
- Olsson et al. — *In-context Learning and Induction Heads*, Anthropic 2022.
- Bricken et al. — *Towards Monosemanticity*, Anthropic 2023.
- Templeton et al. — *Scaling Monosemanticity*, Anthropic 2024.
- Conmy et al. — *Towards Automated Circuit Discovery*, NeurIPS 2023.

8.13 Lab (4 hrs)

1. Install TransformerLens. Load GPT-2 Small.
2. Reproduce the induction-head visualization on a simple `A B A B` pattern.
3. Run Logit Lens through the layers for a factual prompt; visualize trajectory.
4. (Stretch) Use a public SAE (Neuronpedia) — find a feature that activates on "Python code" and ablate it to see effect on next-token predictions.

8.14 Quiz

1. Why is raw attention often not a faithful explanation?
2. Describe activation patching in one paragraph.
3. What's the difference between Logit Lens and Tuned Lens?
4. State the superposition hypothesis.
5. Why are SAEs a promising path for monosemanticity?

XAI Module 9 — Mechanistic Interpretability

> ***Goal***: *Understand the program of reverse-engineering neural networks into human-understandable algorithms — the most active frontier of interpretability research.*

9.1 The mech-interp thesis

> *Neural networks learn algorithms. Those algorithms can, in principle, be reverse-engineered into circuits humans can read.*

If true, this offers something stronger than post-hoc explanation: a **mechanistic understanding** that can be verified, edited, and used for safety.

The thesis is most associated with:

- **Distill.pub Circuits Thread** (2017–2020): Olah, Cammarata, Schubert, Carter, Voss, Goh, Mordvintsev.
- **Anthropic Mechanistic Interpretability team** (2021–present).
- **Neel Nanda** (Google DeepMind / Anthropic): popularized the program with TransformerLens and educational write-ups.

9.2 Why mech-interp matters for AI safety

If you can read what a model is computing, you can:

- Detect deceptive alignment.
- Find dangerous capabilities before deployment.
- Verify whether a model "knows it's lying".
- Edit knowledge / behavior surgically.

These are why Anthropic, DeepMind, OpenAI invest heavily in interpretability research.

9.3 The Distill Circuits Thread — vision

Olah et al. analyzed InceptionV1 unit by unit.

Findings

- **Curve detectors** in early layers — clean equivariant features.
- **High-low frequency detectors** — distinguish foreground/background.
- **Multimodal neurons** (Goh et al., 2021) — single neurons that fire for both an image *and* the text of a concept (e.g., "Spider-Man" neuron).
- **Polysemantic neurons** — one neuron responding to multiple unrelated concepts → superposition.
- **Equivariance** — networks learn rotated/translated copies of similar filters.

These are direct evidence that **interpretable structure exists**.

9.4 A Mathematical Framework for Transformer Circuits (Elhage et al., 2021)

Decompose transformer behavior into:

- **Residual stream** as the communication medium.
- **Attention heads** as read/write operations.
- **MLP layers** as memory / nonlinear gates.
- **QK and OV circuits** as separable read- vs write-direction operators.

This gave the field a vocabulary and made later work (induction heads, ROME) possible.

9.5 Induction heads — Olsson et al., 2022

A specific two-layer circuit that implements in-context completion:

- **Previous-token head** at layer L: copies token T_1 to position of T_2.
- **Induction head** at layer L+1: looks back for similar pattern, copies the continuation.

Together these produce: `[T₁ T₂] ... [T₁] → predict T₂`.

These heads appear **across model scales** and explain a phase transition in in-context learning ability.

9.6 ROME / MEMIT — knowledge editing

Meng et al. (NeurIPS 2022) found that factual associations like "Eiffel Tower is in Paris" are stored in specific MLP layers. ROME edits them via rank-one updates to MLP weights.

Followups:

- **MEMIT** — mass editing (many facts at once).
- **GRACE** — robust against rollback.
- **ICE** — in-context editing without weight changes.

Used in **alignment research** to test whether edited knowledge generalizes (often it doesn't, perfectly).

9.7 Automatic circuit discovery

Manual circuit discovery is slow. Recent work automates it:

- **ACDC** (Conmy et al., NeurIPS 2023) — Automated Circuit DisCovery via edge ablation.
- **EAP / EAP-IG** — Edge Attribution Patching with Integrated Gradients.
- **Path Patching** — measures direct causal paths.

These tools take a hypothesized task (e.g., "indirect object identification") and produce the minimal circuit responsible.

9.8 Indirect Object Identification (IOI) — Wang et al. 2022

Classic case study: GPT-2 Small correctly completes "When John and Mary went to the store, John gave a drink to ___" with "Mary".

Discovered circuit:

- Duplicate-token heads
- S-inhibition heads
- Name mover heads
- Backup name mover heads (redundancy!)

The redundancy explains why simple ablation studies can mislead.

9.9 Superposition and SAEs (recap)

The superposition hypothesis: networks pack more features than they have neurons, in **interfering** directions.

SAEs (covered in Module 8) decompose the residual stream into ~tens of thousands of sparse, near-monosemantic features. This may be the **technical breakthrough** that unblocks mech-interp at LLM scale.

Recent: **Gemma Scope** (Google DeepMind, 2024) — SAEs on every layer of Gemma 2; **SAELens, Neuronpedia, GoodFire** — tooling.

9.10 Causal analysis with abstractions

Geiger, Lu, Icard, Potts (NeurIPS 2021, ICML 2023) — **Causal Abstraction**: hypothesize a human-readable algorithm, then verify by intervention that the model implements it.

Formalizes "the model computes X" as a causal claim that can be falsified.

9.11 Limits of mech-interp

- **Scaling** — most case studies are GPT-2 Small or smaller. Big models are mostly opaque.
- **Polysemanticity / superposition** — even with SAEs, decomposition is imperfect.
- **Compositional behavior** — circuits found for one task may not compose.
- **Reproducibility** — many findings are checkpoint-specific.
- **Selection bias** — researchers report wins; we don't know how often circuits failed to be found.

Be honest about these in any paper you write.

9.12 How to start a mech-interp research project

1. Pick a small, well-defined behavior (1-token completion task).
2. Pick a small model (GPT-2 Small, Pythia 70M–410M).
3. Use TransformerLens. Run ACDC or EAP to find a candidate circuit.
4. Validate via causal interventions (ablation, patching).
5. Write up with diagrams + interactive demos.

The bar to enter is unusually low; the bar to make a *general* claim is high.

9.13 Reading list

- Olah, Mordvintsev, Schubert — *Feature Visualization*, Distill 2017.
- Olah et al. — *Zoom In: An Introduction to Circuits*, Distill 2020.
- Goh et al. — *Multimodal Neurons in Artificial Neural Networks*, Distill 2021.
- Elhage et al. — *A Mathematical Framework for Transformer Circuits*, Anthropic 2021.
- Olsson et al. — *In-Context Learning and Induction Heads*, Anthropic 2022.
- Wang et al. — *Interpretability in the Wild: A Circuit for Indirect Object Identification*, 2022.
- Meng et al. — *Locating and Editing Factual Associations in GPT (ROME)*, NeurIPS 2022.
- Conmy et al. — *Towards Automated Circuit Discovery for Mechanistic Interpretability (ACDC)*, NeurIPS 2023.
- Bricken et al. — *Towards Monosemanticity*, Anthropic 2023.
- Templeton et al. — *Scaling Monosemanticity*, Anthropic 2024.
- Geiger et al. — *Inducing Causal Structure for Interpretable Neural Networks*, ICML 2022.
- Nanda — *A Comprehensive Mechanistic Interpretability Explainer*, neelnanda.io.

9.14 Lab (4 hrs)

1. Install TransformerLens + GPT-2 Small.
2. Reproduce the IOI prompt: "When John and Mary went to the store, John gave a drink to ___".
3. Identify name-mover heads via attention patching. Show that ablating them drops accuracy from ~80% to ~20%.
4. Visualize one head's attention pattern.

9.15 Quiz

1. Why is superposition a problem for naive neuron-level interpretation?
2. State the difference between activation patching and gradient-based attribution.
3. What is an induction head and why does it matter for in-context learning?
4. Describe one finding from the IOI circuit paper that complicates naive ablation.
5. Why might SAEs be the practical breakthrough for LLM interpretability?

Unit 5

Evaluation, Human-Centered XAI, Applications and Frontiers

Required Contact Hours: **12**

- Faithfulness, Plausibility, Robustness; Deletion AUC, ROAR
- Sanity checks (Adebayo), Quantus, ERASER
- Adversarial XAI, Fairwashing, Disagreement problem
- Human-Centered XAI — Miller, trust calibration
- Medical, Financial, Legal XAI; LLM citation grounding
- Research frontiers — SAEs at scale, ZKML, Causal XAI

The chapters that follow expand each topic above with theory, worked examples, reading lists, hands-on lab guidance, and end-of-chapter self-assessment questions.

XAI Module 10 — Evaluating Explanations

> ***Goal***: *A paper that introduces an explanation method is only as good as its evaluation. Master the metrics, sanity checks, and adversarial tests.*

10.1 Three lenses (revisited)

Lens	Question	Cost
Faithfulness	Does the explanation reflect what the model actually computes?	Low–Medium
Plausibility	Does it make sense to humans?	High (user studies)
Robustness	Does it change under tiny input perturbations?	Low

A method can be plausible but unfaithful (Slack et al. attack), or faithful but not human-aligned (mech interp circuits).

10.2 Faithfulness metrics

Deletion / Insertion (Petsiuk, Das, Saenko, 2018 — RISE paper)

```
deletion:  remove top-k most-attributed features → measure prediction drop
           area-under-curve (AUC) = lower is better (explanation captures important feat

insertion: add top-k from blank baseline → measure prediction rise
           AUC = higher is better
```

ROAR (RemOve And Retrain) — Hooker et al., NeurIPS 2019

Remove top features, **retrain** the model, then measure accuracy. Tests whether the attributed features were actually informative — not just "the model uses them now".

Comprehensiveness / Sufficiency — DeYoung et al. 2020 (ERASER benchmark)

- **Comprehensiveness** = drop in prediction after removing the explanation rationale.
- **Sufficiency** = prediction when *only* the rationale is kept.

A good explanation has high comprehensiveness *and* low sufficiency gap.

Infidelity (Yeh et al., NeurIPS 2019)

```
INFD(g, f, x) = E_I [ (I^T g(x) - (f(x) - f(x - I)))² ]
```

Where `I` is a perturbation vector. Measures expected difference between attributions and actual prediction change under perturbations.

AOPC (Area Over Perturbation Curve)

Same family as deletion / insertion AUC.

10.3 Sanity checks (Adebayo et al., NeurIPS 2018)

Two tests every saliency method should pass:

Model parameter randomization

Re-initialize the top layers downward. A faithful saliency must change.

Data randomization

Train the model on randomly-relabeled data. A faithful saliency must produce noticeably different attributions than the correctly-trained model.

Methods that fail: Guided Backprop, Guided Grad-CAM. Their outputs are too similar across these manipulations to count as model-specific explanations.

10.4 Robustness / Stability

Local stability (Alvarez-Melis & Jaakkola, 2018)

```
Stability(x) = sup_{x' ~ x} ‖g(x) - g(x')‖ / ‖x - x'‖
```

For nearby `x'`, the explanation `g` shouldn't jump.

LIME and SHAP can fail this badly without averaging.

Adversarial XAI

Slack et al. (AIES 2020) constructed a "biased model that hides its bias from LIME and SHAP". Showed that explanation methods themselves can be **gamed**.

Aïvodji et al. (2019) — "Fairwashing" — building a fair-looking explanation for an unfair model.

10.5 Plausibility (human-aligned) metrics

ERASER (DeYoung et al., 2020)

NLP benchmark with human-annotated rationales. Compare explanation tokens to human-marked tokens via IoU / token-F1.

Human studies

- **Simulatability** — can humans predict model output given the explanation? (Hase & Bansal, 2020)

- **Forward simulation** — humans use the explanation to make decisions; measure accuracy / time / trust.
- **Counterfactual simulatability** — given an explanation, can the user predict what would change if they changed feature X?

These are the gold standard. They are expensive and rare.

10.6 Specialized benchmarks

Benchmark	Domain
ERASER	NLP rationales
CLEVR-X	VQA explanations
GRADELLM / FActScore	LLM hallucination grounding
OpenXAI	Tabular feature attribution standard
Quantus	Comprehensive XAI metric library (Hedström et al., 2023)
Funke et al. BAM	Benchmark for Attribution Methods (synthetic ground truth)

Use **Quantus** for any new tabular/vision XAI work — it implements 30+ metrics.

10.7 The "disagreement problem" revisited (Krishna et al. 2022)

Empirically:

- Different methods rank features differently.
- Rank correlation across (LIME, SHAP, IG, SmoothGrad) often 0.3–0.6.
- Practitioners pick the explanation that confirms their priors.

Practical mitigation: report multiple methods + consistency metrics; pre-register the explanation method you'll use, before training.

10.8 What to report in a new XAI paper

A minimum checklist:

- [ ] At least one faithfulness metric (deletion/insertion AUC, ROAR, or Infidelity).
- [ ] Sanity checks (Adebayo).
- [ ] Stability metric.
- [ ] Comparison to ≥ 3 baselines.
- [ ] Disagreement analysis with at least 2 alternative methods.
- [ ] If applicable, plausibility metric (ERASER overlap, human study).
- [ ] Computational cost.

Without these, reviewers will ask.

10.9 Reading list

- Adebayo et al. — *Sanity Checks for Saliency Maps*, NeurIPS 2018.
- Hooker et al. — *A Benchmark for Interpretability Methods in Deep Neural Networks (ROAR)*, NeurIPS 2019.
- Yeh, Hsieh, Suggala, Inouye, Ravikumar — *On the (In)fidelity and Sensitivity of Explanations*, NeurIPS 2019.
- Petsiuk, Das, Saenko — *RISE: Randomized Input Sampling for Explanation*, BMVC 2018.
- DeYoung et al. — *ERASER: A Benchmark to Evaluate Rationalized NLP Models*, ACL 2020.
- Slack, Hilgard, Jia, Singh, Lakkaraju — *Fooling LIME and SHAP*, AIES 2020.
- Aïvodji et al. — *Fairwashing: the risk of rationalization*, ICML 2019.
- Krishna et al. — *The Disagreement Problem in Explainable Machine Learning*, 2022.
- Hedström et al. — *Quantus: An Explainable AI Toolkit for Responsible Evaluation*, JMLR 2023.

10.10 Lab (2 hrs)

1. Take last week's saliency methods.
2. Compute deletion AUC and insertion AUC for each.
3. Run Adebayo's model randomization sanity check.
4. Use Quantus to compute Infidelity & Stability.
5. Tabulate results across methods. Which is most faithful? Which is most stable?

10.11 Quiz

1. Why is deletion AUC a faithfulness metric but not a plausibility metric?
2. State the ROAR procedure and what it adds over simple deletion.
3. What did Slack et al. demonstrate, and what does it imply for compliance?
4. Define comprehensiveness and sufficiency from ERASER.
5. Why is the disagreement problem dangerous in regulated industries?

XAI Module 11 — Human-Centered XAI

> ***Goal**: Understand explanations as a **human-computer interaction** problem. The best metric is "did the user understand and act correctly", not "does the heatmap look pretty".*

11.1 The HCI turn

Until ~2018, XAI papers measured success by mathematical properties. Then a wave of HCI researchers (Miller, Hoffman, Mueller, Wang, Lakkaraju, Wattenberg, Viégas) shifted the question:

> *Do explanations actually help users make better decisions?*

The empirical answer is: **often, no**.

11.2 Miller (2019) — what explanation actually is

Explanation in Artificial Intelligence: Insights from the Social Sciences — Artificial Intelligence Journal.

Key findings from 250+ social-science papers on explanation:

1. **Explanations are contrastive** — people ask "why P rather than Q?" not "why P?"
2. **Explanations are selected** — humans don't want every cause, only the most relevant ones.
3. **Explanations are social** — they're a conversation, not a one-shot output.
4. **Probabilities don't help much** — people want causal stories.
5. **People are biased** — they prefer simple, plausible, generalizable explanations even when those are wrong.

Implication: faithful, complete explanations may actively confuse users. Good XAI is selection + framing, not raw attribution dumps.

11.3 Cognitive load & explanation overload

Studies (Bansal et al. 2021, Poursabzi-Sangdeh et al. 2021):

- Showing more features = more confusion.
- Numerical attributions (SHAP values) are misread by non-experts.
- Highlighting *one* feature outperforms showing all features.

Best practice: limit to top-3 to top-5 features for non-expert users. Use plain-language framing ("This loan was likely denied because of low income and short credit history").

11.4 Trust calibration

The goal of an explanation is not "more trust" — it is **appropriately calibrated trust**.

- **Over-reliance** — user trusts wrong answers because the explanation looked convincing.

- **Under-reliance** — user ignores correct answers because of vague unease.

Vasconcelos et al. (CSCW 2022): explanations often **increase reliance regardless of model correctness** — a problem.

Bussone, Stumpf, O'Sullivan (2015) — even nonsensical explanations can increase trust ("automation bias").

11.5 Anchoring & confirmation effects

When an explanation aligns with the user's prior, they adopt the model's prediction. When it contradicts, they often dismiss the model. This is **anchoring**, not understanding.

Mitigation: present alternative hypotheses, not single explanations.

11.6 The "AI-assisted decision-making" framing

Modern HCI–XAI research treats the unit of analysis as the **human-AI team**, not the model alone.

Key questions:

- When should AI defer to humans?
- When should AI override?
- How should AI communicate uncertainty?
- How do we design explanations that improve *team* accuracy, not just model accuracy?

Read: Bansal et al., *"Does the Whole Exceed its Parts?"* (CHI 2021).

11.7 Explanation interfaces

Form	Example product
Local feature highlight	Loan denial reasons
Counterfactual ("if X were different")	Credit improvement coach
Conversational	LLM-grounded "Why did you say that?"
Visualization (PDP, heatmap)	Model debugging dashboards
Concept-based	Medical imaging tools
Prototype / example	Recommender "because you liked X"

The *form* should match the user's mental model. A radiologist wants concepts; a borrower wants counterfactuals.

11.8 Designing a user study for XAI

A minimal protocol:

1. **Task**: real domain task (e.g., loan approval).
2. **Conditions**: no AI / AI-only / AI + explanation type 1 / type 2 / ...
3. **Metrics**: accuracy, time, calibration, trust survey, NASA-TLX (workload).
4. **Participants**: ≥ 30 per condition; domain experts when possible.
5. **Pre-register** hypotheses and analysis plan.
6. **Report** effect sizes, not just p-values.

Most XAI papers skip the user study. The ones that include one tend to overturn the assumption that the "better" explanation is more useful.

11.9 Fairness, accountability, transparency (FAccT)

XAI and fairness intersect but are not the same.

- An explanation may *reveal* unfairness (e.g., heavy weight on a proxy for race).
- An explanation can also *launder* unfairness (Aïvodji 2019, *fairwashing*).

For fairness, use dedicated tools (Fairlearn, AI Fairness 360, Aequitas) **alongside** XAI.

11.10 LLM explanations and "self-explanations"

LLMs can produce natural-language rationales. Beware:

- **Hallucinated rationales** — Turpin et al. (2023) showed CoT explanations can be unfaithful — the model decided early, then rationalized.
- **Sycophancy** — model gives the user the explanation they want.
- **Plausible-sounding mistakes** — the very fluency makes errors more dangerous.

Best practice: verify with non-LLM methods (probes, SAEs, citation-grounded outputs).

11.11 Reading list

- Miller — *Explanation in Artificial Intelligence: Insights from the Social Sciences*, AI Journal 2019.
- Hoffman, Mueller, Klein, Litman — *Metrics for Explainable AI: Challenges and Prospects*, 2018.
- Poursabzi-Sangdeh et al. — *Manipulating and Measuring Model Interpretability*, CHI 2021.
- Bansal et al. — *Does the Whole Exceed its Parts? The Effect of AI Explanations on Complementary Team Performance*, CHI 2021.
- Vasconcelos et al. — *Explanations Can Reduce Overreliance on AI Systems During Decision-Making*, CSCW 2022.
- Bussone et al. — *The Role of Explanations on Trust and Reliance in Clinical Decision Support Systems*, ICHI 2015.
- Turpin et al. — *Language Models Don't Always Say What They Think*, NeurIPS 2023.
- ACM FAccT proceedings — annual.

11.12 Lab (2 hrs)

1. Pick a tabular classifier you trained earlier.

2. Design a 3-condition user study (no-explanation / SHAP / counterfactual).
3. Recruit 6 friends; have each make 20 decisions in each condition.
4. Measure accuracy, time, self-reported confidence.
5. Write a 1-page reflection on what you'd improve.

11.13 Quiz

1. State Miller's four key findings about how humans explain.
2. What is over-reliance and how can XAI cause it?
3. Why might counterfactual explanations be more actionable than SHAP for an end user?
4. What did Turpin et al. show about CoT explanations?
5. Why is faithfulness without plausibility insufficient for end-user XAI?

XAI Module 12 — XAI in Domains: Medical, Finance, Law, NLP

> ***Goal***: *Translate generic XAI methods into domain-specific value (and risk).*

12.1 Medical XAI

Why it matters

- High-stakes decisions
- Clinicians need to validate model reasoning before acting
- Regulatory: FDA SaMD, EU MDR, India CDSCO classify ML-based devices

Common methods by modality

Modality	XAI methods
Radiology (CXR, CT, MRI)	Grad-CAM, LRP, ProtoPNet, BagNet
Pathology (WSI)	Attention-MIL highlights, concept-based
EHR / tabular	SHAP, EBM, GAM
Genomics	DeepLIFT, IG with biological priors
Clinical NLP	Rationale-based, Anchors, IG on tokens

Pitfalls

- **Shortcut learning** — model uses scanner artifacts, image markers, hospital tags (DeGrave et al. 2021 on COVID-19 CXR models).
- **Saliency ≠ diagnosis** — heatmap on lung doesn't mean "consolidation"; could be "rib".
- **Concept misalignment** — concepts defined by ML researchers may not match clinical vocabulary.

Best practices

- Always include an **out-of-distribution test** with a different hospital / scanner.
- Combine pixel attribution with **concept-based** (TCAV, ProtoPNet) for clinical translation.
- Report calibration alongside performance.
- Engage clinicians in evaluation (DECIDE-AI, CONSORT-AI reporting standards).

Key papers

- Caruana et al. — *Intelligible Models for HealthCare*, KDD 2015 (pneumonia EHR example, "asthma is protective" finding — classic shortcut).
- DeGrave, Janizek, Lee — *AI for radiographic COVID-19 detection selects shortcuts over signal*, Nature MI 2021.
- Rajpurkar et al. — CheXpert, CheXNet papers.

- Tonekaboni et al. — *What Clinicians Want: Contextualizing Explainable Machine Learning for Clinical End Use*, MLHC 2019.

12.2 Financial XAI

Why it matters

- Credit decisions are regulated (US ECOA / Fair Lending; EU AI Act high-risk; India RBI guidelines).
- Adverse-action notices legally require reasons.
- Algorithmic trading needs post-mortem analysis.

Common methods

- **SHAP** (industry standard for tabular credit models)
- **Counterfactuals / DiCE** for adverse-action / recourse
- **PDP / ALE** for model debugging
- **EBM** (Microsoft InterpretML) as a credit-modeling baseline
- **Fairness toolkits**: Aequitas, Fairlearn

Pitfalls

- **Proxy features** for protected attributes (zipcode → race).
- **Disparate impact** invisible in SHAP rankings.
- **Adversarial robustness** — gaming the model is profitable.
- **Distribution shift** — credit models built pre-COVID failed badly during the pandemic.

Regulatory landscape (2026)

- US: CFPB / OCC enforce model risk management (SR 11-7).
- EU: AI Act — credit scoring is "high risk", requires logging, explanation, human oversight.
- India: RBI guidelines on AI/ML in financial services.

12.3 Legal XAI

Use cases

- Predictive policing (controversial)
- Recidivism prediction (COMPAS)
- Bail / sentencing risk scores
- Judicial decision prediction (academic, not deployed)

Cautions

- ProPublica 2016 — *Machine Bias* — found COMPAS racially biased. Northpointe disputed. Sparked the **fair prediction debate** (Chouldechova 2017, Kleinberg et al. 2016: impossibility theorem — you can't satisfy multiple fairness criteria simultaneously).
- Rudin (2019) — explicit case for interpretable risk scores over black-box ones in criminal justice.

Best practice

- Use **transparent risk scores** (e.g., 2HELPS2B-style integer-coefficient models).
- Document features, weights, and provenance.
- Audit subgroup performance.

12.4 NLP / LLM XAI

Methods

- Saliency / IG over tokens
- Attention rollout, Chefer-style attribution
- Rationale extraction (ERASER benchmark)
- Influence functions, Datamodels
- Mechanistic interpretability (SAEs, circuits)
- Citation grounding (RAG attribution)

Hot 2024–2026 topics

- **Citation-grounded LLM outputs** (Anthropic Claude Citations, Perplexity sources, OpenAI references)
- **Hallucination detection** via self-consistency, SAE features
- **Sycophancy explanations**
- **Steering vectors / activation engineering** as interpretability + control
- **Constitutional AI / RLHF interpretability**

Open problems

- Faithfulness of natural-language self-explanations (Turpin et al. 2023).
- Cross-lingual interpretability.
- Tool-use explanation in agentic LLMs.

12.5 Computer vision in industry

Application	XAI need
Autonomous driving	Debug failure modes; counterfactual scenes
Manufacturing defect detection	Validate spurious cues
Retail (cashierless)	Audit demographic disparity
Satellite imagery / agriculture	Concept-based features

12.6 Recommender systems

- Show *why* an item was recommended ("Because you watched X").
- Inverse / counterfactual recommendations.
- Methods: PIRL (Persuasive Interpretable Recommendation), counterfactual recommender literature.

12.7 Education

- Adaptive learning systems: explain *why* a concept was recommended.
- AI tutors: rationale-based explanations of answers.
- Diagnostic models for student misconceptions.

12.8 Cybersecurity

- Anomaly detection systems should justify flagged events.
- SHAP/LIME over network features.
- Adversarial-aware explanation (the system being attacked also attacks the explanation).

12.9 Climate / scientific ML

- ML on climate models: which features drive forecast?
- Physics-informed XAI (Toms et al. 2020).
- Concept-aligned attribution for scientific reasoning.

12.10 Domain-XAI checklist (universal)

For any deployment:

- [ ] Stakeholder identified
- [ ] Explanation format matched to stakeholder
- [ ] Faithfulness validated
- [ ] Fairness audit performed independently
- [ ] Shortcut/spurious correlation test passed
- [ ] OOD performance characterized
- [ ] User-facing language tested with domain experts
- [ ] Logging + audit trail in place

12.11 Reading list

- Caruana et al. — *Intelligible Models for HealthCare*, KDD 2015.
- DeGrave, Janizek, Lee — *AI for radiographic COVID-19 detection selects shortcuts*, Nature MI 2021.
- Tonekaboni et al. — *What Clinicians Want*, MLHC 2019.
- Bhatt et al. — *Explainable Machine Learning in Deployment*, FAccT 2020.
- Chouldechova — *Fair Prediction with Disparate Impact*, FAT* 2017.
- Kleinberg, Mullainathan, Raghavan — *Inherent Trade-Offs in the Fair Determination of Risk Scores*, 2016.
- Rudin et al. — *Interpretable Machine Learning: Fundamental Principles and 10 Grand Challenges*, Stat Sci 2022.
- Turpin et al. — *Language Models Don't Always Say What They Think*, NeurIPS 2023.

12.12 Lab (3 hrs)

Pick *one* domain:

- Build a small model (e.g., MIMIC-III mortality / German Credit / IMDB sentiment).
- Apply 2 XAI methods.
- Critique results against domain norms.
- Write a 2-page "would you deploy?" memo.

12.13 Quiz

1. Why did the COMPAS debate produce an impossibility theorem?
2. What was Caruana's "asthma is protective" finding and what does it teach?
3. Why are concept-based methods better than saliency for clinician communication?
4. Name two ways shortcut learning shows up in medical imaging.
5. Why might LLM self-explanations be unfaithful?

XAI Module 13 — Research Frontiers (2025–2027)

> ***Goal***: *Identify the active research clusters and where a Master's student can make a tractable contribution.*

13.1 The XAI research map (2026)

```
                  ┌─────────────────────────────────────────┐
                  │   MECHANISTIC INTERPRETABILITY (LLMs)    │
                  │  SAEs · Circuits · Probing · Steering   │
                  └─────────────────────────────────────────┘
     ┌───────────────────────────┐   ┌───────────────────────────────┐
     │ CAUSAL & COUNTERFACTUAL   │   │   EVALUATION & BENCHMARKING   │
     │  Algorithmic recourse     │   │ Quantus · ERASER · OpenXAI    │
     │  Causal abstractions      │   │ Adversarial XAI · disagree.   │
     └───────────────────────────┘   └───────────────────────────────┘
     ┌───────────────────────────┐   ┌───────────────────────────────┐
     │  HUMAN-AI INTERACTION     │   │ DOMAIN: MED / FIN / LAW       │
     │  Calibrated trust         │   │ Clinical translation          │
     │  Team performance         │   │ Fairwashing                   │
     └───────────────────────────┘   └───────────────────────────────┘
     ┌──────────────────────────────────────────────────────────────┐
     │  EMERGING: ZKML · VERIFIABLE EXPLANATIONS · DECENTRALIZED ML │
     └──────────────────────────────────────────────────────────────┘
```

13.2 Hot subareas

1. Mechanistic interpretability at scale

- SAEs on 100B+ parameter models.
- Automated circuit discovery (ACDC, EAP-IG).
- Cross-model universality of circuits.
- Practical alignment use cases (deception, jailbreak detection).

2. Faithful CoT and chain-of-thought interpretability

- Turpin et al. showed CoT may be unfaithful.
- Open: how to *train* models with faithful reasoning?
- Process supervision (Lightman et al. 2023 OpenAI) is a partial answer.

3. Multimodal & agent interpretability

- VLMs (CLIP, BLIP, Gemini): cross-modal feature attribution.
- LLM agents: explaining tool-use sequences.
- Reward model interpretability (what does the RLHF reward model actually reward?).

4. Causal XAI

- Counterfactuals with causal graphs.
- Algorithmic recourse under uncertain causal knowledge.
- Causal discovery from observational data + XAI integration.

5. Adversarial XAI

- Robust explanations that survive adversarial perturbations.
- Detecting fairwashing.
- "Honest" model architectures that can't lie to explainers.

6. XAI for foundation models / diffusion models

- Interpreting CLIP-like models.
- Concept activation for image generation control.
- Diffusion model interpretability via score landscapes.

7. Probing and feature discovery

- Sparse probing.
- Linear representation hypothesis tests.
- Geometry of representations (Park et al. 2023, *Linear Representation Hypothesis*).

8. Memorization vs generalization

- Influence functions at scale (Datamodels, TRAK).
- Membership inference and unlearning explanations.

9. Evaluation crisis

- Most explanation methods evaluated on toy benchmarks.
- Need standardized, application-grounded evals.
- Quantus, OpenXAI lead this.

10. XAI + Privacy

- Differentially-private explanations.
- ZK-proven explanations (ZK + XAI).

11. XAI + Fairness

- Explanations as fairness audits.
- Fairwashing detection.
- Subgroup-aware explanations.

12. Verifiable / On-Chain ML (zkML)

- Prove inference was correct + model unchanged.
- See Intersection module.

13.3 Tractable Master's-thesis-scale topics

Topic	Why tractable
Empirical disagreement study on a new domain (legal text, audio)	Reuse existing tools
SAE feature analysis on a specific small model	TransformerLens makes setup fast
User study comparing 2 explanation forms on a real task	High novelty, modest infrastructure
Counterfactual quality for credit data under causal constraints	Data is available, methods extant
Reproducibility study of 5 saliency methods	Always publishable as a workshop paper
zkML proof of a tiny network's inference + interpretability	Crossroads area, sparse competition

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13.4 Venues for XAI research

Venue	Type
NeurIPS, ICML, ICLR	Tier-1 ML
AAAI, IJCAI	Broader AI
ACL, EMNLP, NAACL	NLP, XAI for language
CVPR, ICCV, ECCV	Vision XAI
FAccT	Fairness, accountability, transparency
CHI, CSCW, UIST	HCI, user studies
AIES (AAAI/ACM)	AI ethics
MLHC	Healthcare ML
JMLR, Nature MI, Distill (defunct but archive)	Journals

For first papers: **XAI4CV, Re-XAI, BlackBoxNLP** workshops are friendly entry points.

13.5 How to identify a good open problem (workflow)

1. Pick a frontier (e.g., faithful CoT).
2. Read the 10 most-cited papers in that area from the past 2 years.
3. Tabulate **claims**, **assumptions**, **limitations** of each.
4. Find:
    - A claim with weak empirical support → reproduce + extend.
    - An assumption no one tested → test it.
    - A limitation that has a small-scope fix.
5. Run a 2-week pilot. If it works, commit to a 3-month project.

13.6 Reading list — recent must-reads

- Bricken et al. — *Towards Monosemanticity*, Anthropic 2023.
- Templeton et al. — *Scaling Monosemanticity*, Anthropic 2024.
- Conmy et al. — *Automated Circuit Discovery (ACDC)*, NeurIPS 2023.
- Turpin et al. — *Language Models Don't Always Say What They Think*, NeurIPS 2023.
- Krishna et al. — *The Disagreement Problem in Explainable ML*, 2022.
- Rudin et al. — *Interpretable ML: Fundamental Principles and 10 Grand Challenges*, Statistical Science 2022.
- Park et al. — *The Linear Representation Hypothesis and the Geometry of Large Language Models*, 2023.
- Marks et al. — *Sparse Feature Circuits*, 2024.
- Karimi et al. — *A Survey of Algorithmic Recourse*, ACM CSUR 2022.

13.7 Lab / capstone

Pick **one**:

- Apply Sparse Autoencoders to Pythia-160M on a custom domain (e.g., medical text). Browse and document 50 interesting features.
- Empirical disagreement study: 6 methods × 4 datasets × 3 model classes. Quantify disagreement.
- User study: 30 participants, 2 explanation conditions, 1 domain task. Pre-register.
- Reproduce ACDC's IOI circuit; extend to a different prompt template.

13.8 Quiz

1. Name three SAE-related results from 2023–2024.
2. Why is faithful CoT an unsolved problem?
3. Define "linear representation hypothesis" in one sentence.
4. What makes algorithmic recourse harder than counterfactual explanation?
5. Pick one workshop venue you'd target with a first paper, and justify the fit.

Appendix

Appendix A — XAI × Blockchain Research Intersection

The chapter that follows links this subject to its sister discipline, surveying the small but active research area where the two converge.

Intersection — Verifiable, Explainable, and Decentralized AI

> ***Goal**: This is the research-value module. Where Blockchain meets XAI, several emerging research lines need contributors. A Master's thesis here has unusually high impact-per-effort ratio.*

I.1 Why these two fields converge

Blockchain solves: **verifiability and integrity of computation in adversarial environments.** XAI solves: **understanding and trust of model behavior.**

Together, they tackle the question:

> *Can we know — and prove — what an AI did, why it did it, and that no one tampered with it?*

Three convergence areas:

1. **Verifiable AI** — cryptographic proofs that a model output is correct (ZKML).
2. **Decentralized AI** — model training/inference distributed and incentivized via tokens.
3. **Auditable / Explainable AI** — on-chain logs and explanations for regulatory compliance.

I.2 ZKML — Zero-Knowledge Machine Learning

The problem

A model owner can prove to a verifier that:

- "This output `y` is the result of running model `M` on input `x`."

Optionally hiding:

- The model weights (IP protection)
- The input (privacy)
- Both

Why hard

Neural networks use:

- **Floating-point arithmetic** — ZK circuits work over finite fields → quantization required.
- **Non-linear ops** (ReLU, GELU, softmax) — expensive to express in arithmetic constraints.
- **Large parameter counts** — billions of multiplications → billions of constraints.

Active toolchains (2024–2026)

Tool	Approach

EZKL	Halo2 backend, ONNX-friendly. Quantization + lookup-arg optimizations
Modulus Labs (Remainder)	GKR / sum-check for matrix multiplications
Risc Zero zkVM	General-purpose zkVM, run PyTorch in Rust via tract
Giza	StarkNet + Cairo for on-chain inference
DDKang's zkml	Halo2, focused on ResNet-scale
Inference Labs	Production ZK inference SaaS
Worldcoin	Uses semaphores + ZK for proof of personhood

State of the art

- ResNet-50 inference: provable in ~minutes on modern hardware, ~MB proofs.
- Small CNN MNIST: seconds, KB proofs.
- LLMs (>1B parameters): largely impractical end-to-end today; partial proofs possible.

Why it matters for XAI

A ZK proof of *correct inference* is necessary but not sufficient for trust. The next step: **ZK proofs of explanation**. Examples:

- Prove that a SHAP value reported externally was correctly computed from the model.
- Prove that a counterfactual explanation is genuinely the minimum-cost one.
- Prove that an attention-based "reason" was correctly extracted.

This is **mostly open research as of 2026** — there's room for early entrants.

I.3 On-chain machine learning

Approaches

Approach	Trust assumption
Inference **off-chain**, post **proof on-chain**	ZK or optimistic
Inference **on-chain** in a smart contract	High gas; bounded models only
Inference in **TEE** with attestation	Hardware trust (Intel SGX, AMD SEV)
Inference via **MPC**	Multiple non-colluding parties
Inference with **FHE**	Pure cryptography, very expensive

EVM-based on-chain ML (Modulus Labs' Ocean, ORA) typically posts ZK proofs of small models for verifiable randomness, oracle aggregation, content moderation triggers, etc.

I.4 Decentralized AI infrastructure

Compute marketplaces

Project	Layer
Akash	General Kubernetes compute
Render	GPU rendering + ML inference
io.net	Distributed GPU cluster
Gensyn	Distributed training with verification
Bittensor	Subnet-based incentivized model marketplace
Ritual	AI inference oracle for smart contracts

Trust model varies — some use ZK, some use challenge-response / fraud proofs.

Model marketplaces & reputation

Open problems:

- How do you verify a model's claimed performance?
- How do you reward good models without enabling sybils?
- How do you handle adversarial submissions (model poisoning)?

Bittensor's "Yuma consensus" weights subnet contributions by peer voting — a research-grade mechanism design effort.

I.5 Federated learning + blockchain

Federated learning (FL) trains a model across many devices without centralizing data. Blockchains can:

- Coordinate aggregation rounds.
- Reward participants by stake / contribution.
- Audit updates for poisoning.
- Verify aggregation via ZK / MPC.

Research papers stack: FL + blockchain + differential privacy + XAI.

Trust model: typically a permissioned consortium chain (Hyperledger Fabric, IBC chains). Honest-majority assumption.

I.6 Decentralized identity and AI

Tool	Use
W3C DIDs / Verifiable Credentials	Self-sovereign identity
Soulbound tokens (SBT)	Non-transferable reputation

zkID / Sismo / Worldcoin	Privacy-preserving "uniqueness" proofs
Polygon ID	Selective disclosure of attributes

Why for AI: deepfake / agentic-AI era requires **proof of personhood** to distinguish humans from agents. Worldcoin's iris-scan-based PoP is the most controversial deployment.

I.7 Audit trails: explanations on-chain

For regulatory compliance, you may want:

```
inputs_hash + model_hash + outputs_hash + explanation_hash → on-chain commitment
```

Anyone can later verify the audit trail. Variants:

- **Off-chain explanation, on-chain hash** (cheap, requires data availability).
- **On-chain explanation** (expensive, only for small models or critical decisions).
- **ZK explanation** (prove the explanation was correctly computed without revealing input or model).

Use cases:

- Credit decision explanations stored immutably for regulator audit.
- Healthcare AI outputs with retroactive verifiability.
- Election counting AI with public auditability.

I.8 Adversarial concerns

This intersection inherits **both** fields' attack surfaces:

From blockchain	From XAI
51% attack on verification chain	Adversarial explanations (Slack et al.)
Bridge hacks (cross-chain proofs)	Fairwashing
Oracle manipulation	Manipulated saliency
MEV on inference auctions	CoT unfaithfulness
Smart contract bugs in verifier	Proof-system soundness bugs

New combined risks:

- **Adversarial ZK proofs** — soundness errors in custom circuits.
- **Quantization attacks** — exploit the fp→fixed conversion to skew explanations.
- **Replay of stale model commitments**.
- **Sybil attacks on decentralized model voting**.

I.9 Open research questions

1. **Sub-second ZK proofs of small NN inference + SHAP** — currently minutes; gap is 100×.
2. **Faithful, provable LLM explanations** — combining mech-interp with ZK.
3. **Decentralized model training with verifiable convergence** — beyond FedAvg + Merkle commits.
4. **Privacy-preserving fairness audits** — prove fairness without revealing individual records.
5. **Soulbound reputation for model providers** with non-gameable scoring.
6. **Cross-chain AI oracles** — AI predictions consumed across multiple L1s.
7. **MEV-protected AI inference markets**.
8. **Compliance-grade audit trails** for EU AI Act, with ZK for trade-secret protection.
9. **On-chain explanation registries** that survive model versioning.
10. **Watermarking AI outputs** with blockchain-anchored provenance (deepfake defense).

I.10 Sample research project (Master's-scale)

Title: "Verifiable Feature Attribution for Black-Box Inference: A ZK-SHAP Prototype"

Plan:

1. Train a small (1–5M param) tabular model.
2. Implement KernelSHAP in a ZK-friendly form (fixed-point, batch evaluation).
3. Compile to EZKL / Risc Zero.
4. Generate proofs for ~10 test inputs; measure proof size, prover time, verifier gas.
5. Test soundness via adversarial model (Slack attack adapted) — does the proof catch it?
6. Write up as a workshop paper (target: AFT, IEEE Blockchain, BlackBoxNLP, or zkProof workshop).

This is a clean, novel intersection topic with a 3–6 month timeline.

I.11 Reading list

- Kang, Hashimoto, Stoica, Sun — *zkML and verifiable inference* (various 2023–2024 papers).
- Lee, Kang, Lee, Boneh — *EZKL and Halo2 ZK Inference*, 2023.
- Feng, Yan, Zhao, Lai, Lin — *Cerberus: ZK-SNARK-based Verifiable ML Inference*, 2021.
- Liu, Xie, Zhang — *zkCNN: Zero-Knowledge Proofs for Convolutional Neural Networks*, CCS 2021.
- Modulus Labs — *Cost of Intelligence* benchmark.
- Bittensor whitepaper.
- McMahan et al. — *Communication-Efficient Learning of Deep Networks from Decentralized Data (FedAvg)*, AISTATS 2017.
- Bonawitz et al. — *Practical Secure Aggregation for Privacy-Preserving ML*, CCS 2017.
- Karimi et al. — *Verifiable Federated Learning via ZK proofs*, various 2023.
- W3C — *DID Specification*; *Verifiable Credentials Data Model*.
- *EU AI Act* — Articles on high-risk systems and transparency.

I.12 Lab (5 hrs)

1. Install EZKL.
2. Train a 2-layer MLP on MNIST in PyTorch; export to ONNX.
3. Run `ezkl gen-settings`, `compile-circuit`, `setup`, `gen-witness`, `prove`, `verify`.

4. Time the proving step; measure proof size.
5. Compute a SHAP explanation off-chain; commit its hash on Sepolia.

I.13 Quiz

1. Why are ReLU and softmax expensive in ZK circuits?
2. State three trust models for on-chain ML.
3. How would you prove "this SHAP value was correctly computed" without revealing the model?
4. Why does Bittensor's Yuma consensus need a mechanism-design contribution to remain non-gameable?
5. Pick one of the 10 open research questions and write a 3-sentence research plan.